

Grade 7 Accelerated
Unit 3
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

1. Write the product as a repeated multiplication expression, then write the expression with an exponent.

Product	Repeated Multiplication	Power of the Form a^b
$2^2 \cdot 2^3$	$(2 \cdot 2) \cdot (2 \cdot 2 \cdot 2) = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$	2^5
$4^2 \cdot 4^5$	$(4 \cdot 4) \cdot (4 \cdot 4 \cdot 4 \cdot 4 \cdot 4) = 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4$	4^7

2. A base is the number used as a factor in exponential form. An exponent is the number of times the base occurs as a factor.

3. Evaluate each power then multiply.

$$5^3 \cdot 5^2$$
$$125 \cdot 25 = 3,125$$

4. Apply the product of powers exponent law, then evaluate the power.

$$(-6)^2 \cdot (-6)^7$$
$$(-6)^9$$
$$-10,077,696$$

5. Leeland says that he can evaluate $7^3 \cdot 3^7$ using the product of powers exponent law. Explain to Leeland if he is correct.

He is not because he does not have like bases.

6. Write each power of a product as a repeated multiplication expression and group repeated factors by using the commutative and associative properties. Then, write each expression with an exponent in the form $a^m \cdot b^n$.

Power of a Product	Repeated Multiplication	Group Repeated Factors	Power of the Form $a^m \cdot b^n$
$(2 \cdot 3)^3$	$(2 \cdot 3)(2 \cdot 3)(2 \cdot 3)$	$(2 \cdot 2 \cdot 2)(3 \cdot 3 \cdot 3)$	$2^3 \cdot 3^3$
$(7 \cdot 8)^2$	$(7 \cdot 8)(7 \cdot 8)$	$(7 \cdot 7)(8 \cdot 8)$	$7^2 \cdot 8^2$

7. Evaluate inside the parenthesis, then apply the exponent.

$$\begin{aligned}
 &(-3 \cdot 6)^4 \\
 &(-18)^4 \\
 &104,796
 \end{aligned}$$

8. Apply the power of a product exponent law, then evaluate the power.

$$\begin{aligned}
 &(2 \cdot -9)^3 \\
 &2^3 \cdot (-9)^3 \\
 &8 \cdot (-729) \\
 &-5,832
 \end{aligned}$$

9. Use the power of a power exponent law to simplify the expression.

$$[(5)^4]^4 = \underline{\quad 5^{16} \quad}$$

10. Rewrite the expression by using the laws of exponents. Circle the exponent law used from the list provided. Then, evaluate the exponential expression.

Expression	Exponent Law	Evaluated Expression
$[(2)^5]^2$	• Product of powers law • Power of product law • Power of power law • Identity exponent law	2^{10}